

Research article

Off grid PV/Diesel/Wind/Batteries energy system options for the electrification of isolated regions of Chad

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ABSTRACT

Access to reliable energy is fundamental for the development of any community. The electricity is produced in Chad solely from thermal plants that use fossil fuels, which are not environmentally friendly. In addition, the electrification rate of Chad is less than 11%. This work aims to propose some reliable electrification options for Chad, through hybrid energy systems. To achieve this objective, autonomous hybrid PV/Diesel/Wind/Batteries feasibility to meet the demand of electrical load in isolated regions of Chad is evaluated using HOMER software. The design is done considering three types of daily load profiles in each of the 16 regions that are not yet electrified in Chad; the low, medium and high community load profiles. From the simulation, it was observed that the optimal configurations were: PV/Battery, PV/Diesel/Battery and PV/Wind/Diesel/Battery for various consumers and sites. The COE was found to be in the range of 0.367 and 0.529 US\$/kWh which shows that, the COE of some sites are less than the production cost of energy in Chad (0.400 US\$/kWh) and therefore profitable. Using these hybrid systems, compared to single diesel generator will result in less CO₂ emission per year (between 0 and 15670 kg/year). These results may guide investors and policies makers in the planning and implementation of various optimal feasible options that may be used to increase the electricity access rate of Chad, especially in remote areas.

1. Introduction

Development is related to the electricity access rate. Global demand of energy is growing due to the increasing industrialization and population. The traditional energy resources cannot meet these demands without taking into account the challenges of harmful gas emissions and high fuel life cycle costs [1]. Renewable energy sources are available, inexhaustible and clean to the environment. The quest for sustainable and clean energy is growing globally rapidly due to the increasing world population and the need to reduce CO₂ emissions [2,3]. The access rate to electricity is still very low in developing and underdeveloped countries. This is the case of Chad where the electricity access rate are only 11% and 2% respectively for the urban and rural population [4]. Due to renewable energy sources uncertainties, the combination of energy storage system with sources is a way to increase the system reliability [5].

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The life cycle cost of hybrid Solar/Diesel/storage systems are less expensive than that of a single Diesel generator. Compared to the system using only fossil fuels, with the optimized hybrid energy systems, the CO₂ emissions is reduced by approximately 62% [6]. They provide reliable power and reduce the emissions of greenhouse gases [7]. A hybrid system composed of wind, PV and biomass energy is cost-effective and reliable for sustainable electrification of rural areas with environmental benefits [8]. It was proven to be economical compared to electricity supply to the remote location with only solar systems [9]. The use of micro-grids operating with combined energy sources is increasing as it is a solution for the supply of remote locations where the expansion of the distribution network is not feasible [10]. The solar/battery system was found to be the most economically suitable option for autonomous electricity production in northern Nigeria [11]. The energy cost of a grid-connected system is lower than that of an off-grid system for similar load demands [12]. Hybrid off-grid system is more reliable and cost-effective than single system source for rural electrification [13]. The sustainability assessment noted that preventing the grid from charging the batteries would result in a higher renewable fraction [14]. When the system is integrated into the network, it is more economical than off-grid systems [15].

In this study, the hybrid energy systems are proposed for all the regions that are not yet electrified in Chad. The National Electricity Company (NEC) of Chad produces and distributes the electricity only in 7 of the 23 regions of Chad; meaning that 16 are un-electrified. In many isolated areas of the country, access to electricity remains a problem for the population. The continuity of electricity supply at all times is difficult because in some areas, the population has access to electricity between 6 p.m. and 10 p.m., approximately only 4 h a day. It is essential to explore the abundant potential of the wind and the possibilities of using the wind and solar energy conversion system as sources of electricity with the aim of meeting the energy needs of Chad. Harnessing the wind and solar energy could contribute to sustainable energy development. Chad has high solar potential and therefore conducive to the operation of solar systems [16]. The global solar radiation varies from 4.5 to 6.5 kWh/m²/d. For the wind energy, the speed of calm winds varies from 4 m/s to 9



Fig. 1. Map of study area [21].

m/s from south to north [17]. Our motivation aims to propose hybrid energy systems to resolve the low access rate of electricity in Chad. Most Chadians live in villages without a particular electricity supply system, the implementation of renewable energy is a way to promote development and improve the well-being of the populations [17]. The country being located where the solar and wind potential is high, this asset can increase access rate of electricity.

Even though many previous works related to hybrid energy system sizing are found in the literature, to the best of our knowledge, only four are applied for some sites of Chad [3,18,19,20]. [17] assessed the Grid/PV/Wind hybrid energy system viability to provide electricity in 25 sites of Chad [18]. designed a solar/wind/diesel/batteries for three climatic zones of Chad [19]. investigated the feasibility of solar/wind/diesel/batteries for the supply of energy needs of Amjarass (a town in Chad). However, these previous works applied in Chad have some limitations: In Ref. [17], even though 25 sites of Chad were considered, only one load profile obtained instead for a site in India was used. In Ref. [18], even though four hypothetical load profiles were considered in each of the three climatic zones, the authors supposed that the sites of Faya, Pala and Abeche represent the entire country. In Ref. [20], only one site (Amjarass) with only one hypothetical load profile was considered. The authors of [3] on their part, proposed the hybrid system for only five sites of Chad. The investigation of hybrid energy system feasibility has not yet been studied for most remotes areas of Chad where there is no access to electricity. This paper main objective is to contribute to the electrification of isolated sites in Chad with renewable energy sources. In this work, hybrid energy options are simulated with the HOMER (Hybrid Optimization Model for Electric Renewables) software considering three types of community load profiles for each of the 16 un-electrified regions of Chad. In the rest of the paper, the methodology employed is firstly presented before a discussion on the results obtained. The work ends with a summary on the main findings obtained.

2. Materials and methods

2.1. Study areas

Chad, a country in Central Africa has a surface area of 1284000 km² and N'Djamena as its capital. According to the new administrative division of 2018, the country has 23 regions. The country is limited by six countries: Cameroon, Central African Republic, Niger, Nigeria, Sudan and Libya. It extends from North to South to over 1700 km and from East to West to over 1000 km. The country is characterized by three dominant geological zones with varying rainfall. The Saharan zone extends to approximately 780,000 km² and covers approximately 50% of the country's surface area, the Sahelian zone covers approximately 374,000 km², and the Sudanese zone occupies approximately 130,000 km² (Fig. 1.).

The NEC of Chad produces electricity solely by thermal plant, which is not environmentally friendly. This electricity production system consumes a lot of fuel, is expensive and very polluting [22]. Chad has significant renewable energy potential that may be exploited, such as biomass, wind, solar and hydroelectricity, which are still untapped. Also, the supply of electricity by NEC is limited only in 7 of the 23 regions of Chad; Abeche, Doba, Bongor, Faya-Largeau, Moundou, N'Djamena and Sarh. The others 16 regions of Chad that are not yet electrified are considered in this work. These considered regions are: Mao, Bol, Mongo, Moussoro, Biltine, Gozbeida, Ati, Messenya, Massakory, Amdjarass, Bardai, Fada, Pala, Lai, Amtiman and Koumra.

2.2. Electricity demand assessment

For this work we used three type of community load profile obtained through a survey in Ref. [3]; the low, the medium and the high

Table 1
Estimation of each type of community load profile [3].

High profile consumption						
Device	Lamps	Phone Charger	Television	Ventilator	Fridge	Air conditioner
Power (W)	40	20	200	40	200	1200
Number of device	5	1	2	2	1	1
Total power (W)	200	20	400	80	200	1200
Number of hours/day	6h	2h	4h	8h	8h	4h
Energy consumed per day (Wh)	1200	40	1600	640	1600	4800
Medium profile consumption						
Device	Lamps	Phone Charger	Television	Ventilator	Fridge	Air conditioner
Power (W)	40	20	200	40	200	1200
Number of device	5	1	1	2	1	0
Total power (W)	200	20	200	80	200	0
Number of hours/day	6	2	4	4	4	0
Energy consumed per day (Wh)	1200	40	800	320	800	0
Low profile consumption						
Device	Lamps	Phone Charger	Television	Ventilator	Fridge	Air conditioner
Power (W)	40	20	200	40	200	1200
Number of device	1	1	0	0	0	0
Total power (W)	40	20	0	0	0	0
Number of hours/day	6h	2h	0	0	0	0
Energy consumed per day (Wh)	240	40	0	0	0	0

community consumers. Given the Chadian context, each type of the three consumers can be identified in each of the 16 isolated un-electrified regions of the country. The estimation of each community consumer is presented in Table 1 [3].

Fig. 2 shows the variation in the daily electrical energy demand of the three community sites. We notice that the maximum is reached at 7 p.m. with a value of 144 KW for the high load profile. For the average load, the maximum is reached at 3 p.m. with a value of 159 KW. The maximum low load profile is reached at 4 p.m. with a value of 11 KW. The particularity of these different types of loads is that the consumption is zero from 0 to 7 a.m. in the morning. For the medium and low load profile, the consumption is zero from midnight to 3 p.m. because in the Chadian context during the day, the populations are not at home.

2.3. Resource assessment

In this study, we used the solar irradiation as well as the wind speeds to simulate the hybrid wind/PV/diesel/batteries system feasibility.

2.3.1. Solar potential

For this work, we have used 12 years (from 2010 to 2021) monthly average solar irradiation data downloaded from a NASA website (<https://power.larc.nasa.gov>) for each of the 16 sites. Due to limited space, only the solar irradiation of 3 sites, each representing one geological zone are presented from Figs. 3–5. The Sahelian zone takes into account Mao, Bol, Mongo, Moussoro, Biltine, Gozbeida, Ati, Messenya and Massakory. In this area we notice that the irradiation are high from March to May and the most unfavorable month is December (Fig. 3). The Saharan zone of Chad covers the following regions: Amdjarass, Bardai and Fada. This region has low irradiation in January and high irradiation from April to June as shown in Fig. 4. The Sudanian zone covers the regions of Pala, Lai, Amtiman and Koumra. Fig. 5 shows the solar irradiation data of the Sudanese zone; we note that in the month of April the irradiation is higher and the unfavorable month is the month of December.

2.3.2. Wind potential

The wind speeds considered for this work in the 16 un-electrified regions of Chad, these data are measured at 50 m above the ground. They are shown in Figs. 6–8 each representing one site in the three geological zones. The Sahelian zone shows the least favorable wind speed in September. The wind speeds are very high in the Saharan and Sahelian zone which are in the northern part of the country compared to the Sudanese zone. In the Sudanese zone (the south of the country), the wind speeds are approximately lower.

2.4. Components of hybrid systems studied

In this work the hybrid systems studied are of PV, wind, diesel generators, batteries, and converters. The hybrid energy system is shown in Fig. 9.

Table 2 gives details of the different components with their costs in the Chadian market.

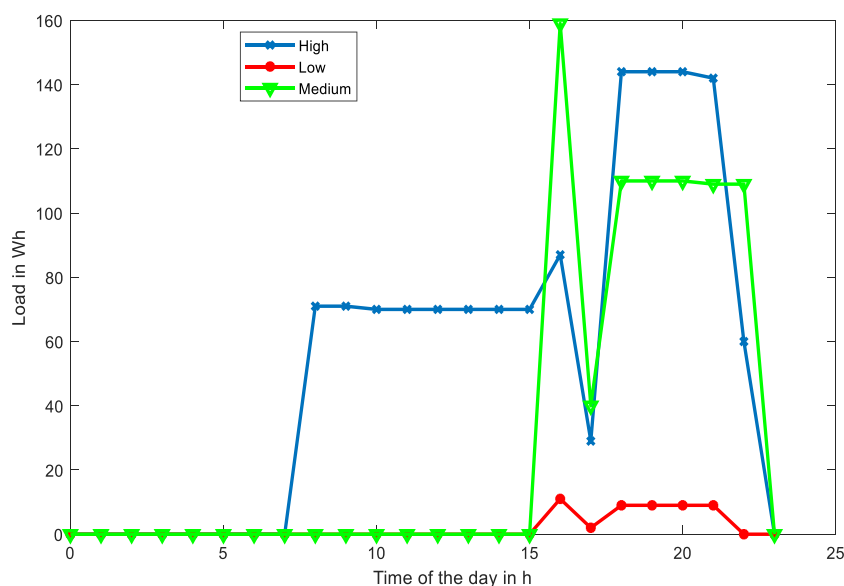


Fig. 2. Load curve of daily electrical demand.

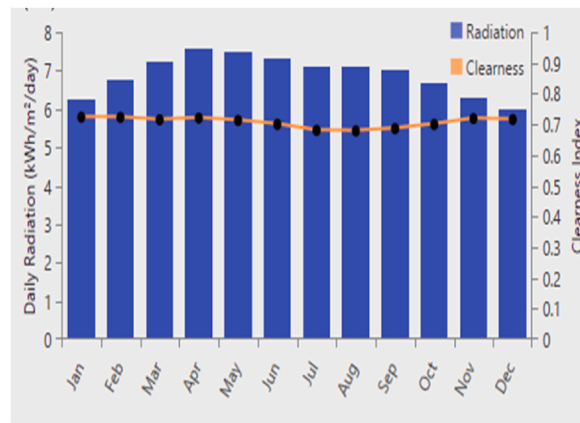


Fig. 3. Solar resources of Biltine (Sahelian zone).

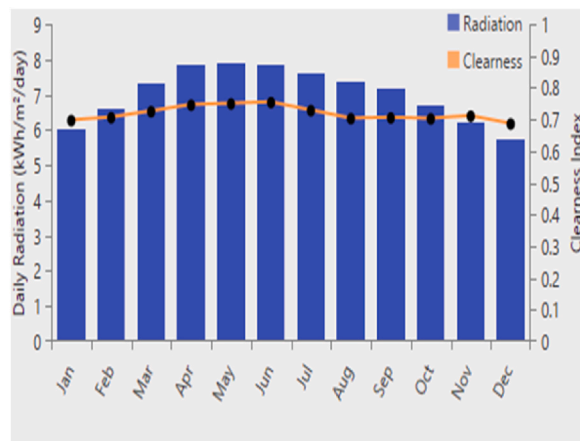


Fig. 4. Solar resources of Amdjarass (Saharan zone).

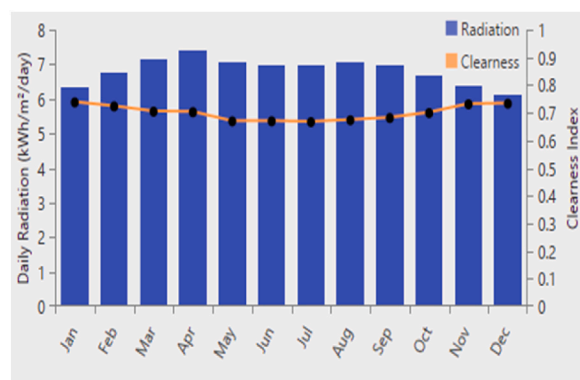


Fig. 5. Solar resources of Pala (Soudanese zone).

2.5. Solar energy

The solar energy produced is given by the expression of equation (1) [23]:

$$P_{pv-out} = P_{pv-rated} \left(G / G_{ref} \right) \left[1 + K_T (T_c - T_{ref}) \right] \quad (1)$$

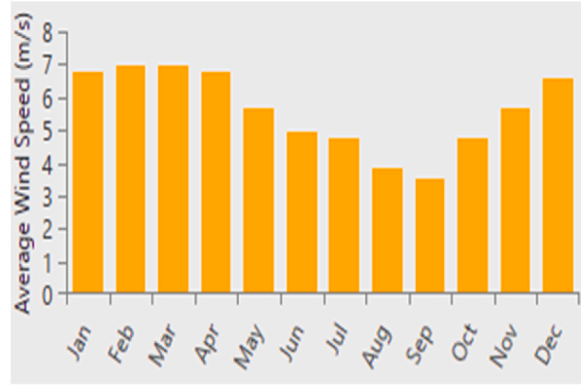


Fig. 6. Wind Speed of Biltine (Sahelian zone).

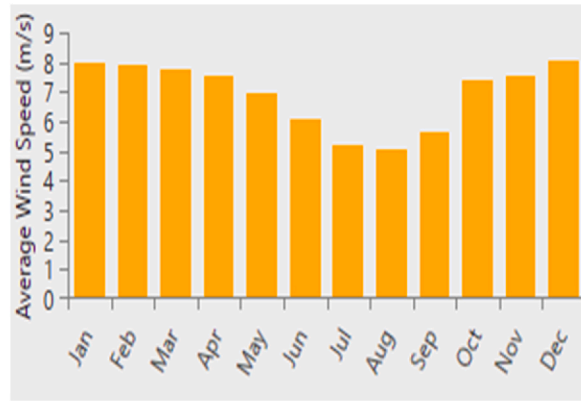


Fig. 7. Wind Speed of Amdjarass (Saharan zone).

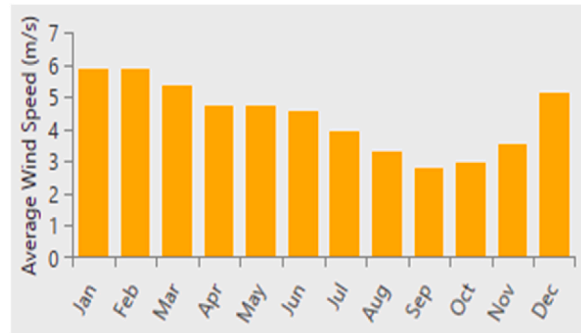


Fig. 8. Wind Speed of Pala (Soudanese zone).

where P_{pv-out} represents the power output of the PV, $P_{pv-rated}$; the PV rated power at reference test condition, G ; the solar radiation (W/m^2), G_{ref} ; the solar radiation at standard temperature condition ($G_{ref} = 1000 W/m^2$), T_{ref} ; the cell temperature at reference conditions ($T_{ref} = 25 ^\circ C$), K_T ; the temperature coefficient of the PV module.

2.6. Wind energy

The output power of wind turbines is determined by equation (2) [24]:

$$P_{WECS} = \frac{1}{2} \rho A V_S^3 C_p(\lambda, \beta) \eta_t \eta_g \quad (2)$$

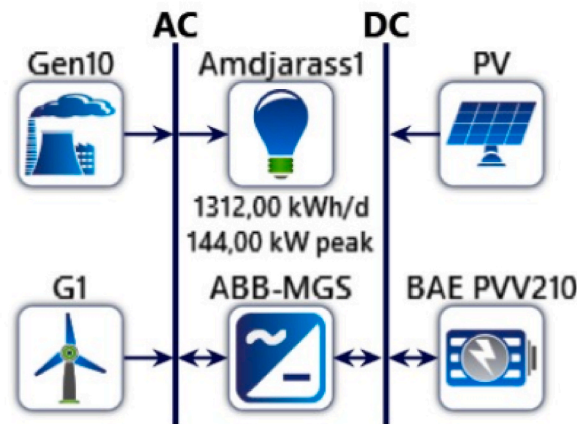


Fig. 9. Considered hybrid system.

Table 2
Components costs [3].

PV System					
Description	Capacity (Kw)	Capital (\$)	Replacement cost (\$)	O&M cost (\$/yr)	Lifetime (years)
Specification	1	270	243	2.70	25
Inverter					
Description	Capacity (Kw)	Capital (\$)	Replacement cost (\$)	O&M cost (\$/yr)	Lifetime (years)
Specification	1	984	886	9.80	15
Battery					
Description	Capacity (Kw)	Capital (\$)	Replacement cost (\$)	O&M cost (\$/yr)	Lifetime (years)
Specification	2.64	321	289	32	18
Wind turbine					
Description	Capacity (Kw)	Capital (\$)	Replacement cost (\$)	O&M cost (\$/yr)	Lifetime (years)
Specification	1	5320	5320	532	20
Generator					
Description	Capacity (Kw)	Capital (\$)	Replacement cost (\$)	O&M cost (\$/yr)	Lifetime (years)
Specification	10	5000	5000	0.30	15000

Where, P_{WECS} is the Output power of WECS; ρ is the air density; A is the area swept by rotor blades; V_s is the velocity of wind; C_p is the performance coefficient of wind turbine; λ is the tip-speed ratio of blade; β is the blade pitch angle; η_t and η_g are respectively the wind turbine and generator efficiency.

2.7. Diesel generator

The diesel generator is used as a backup system, when the renewable resource is not able to meet demand. The expression of the diesel consumption is given by equation (3) [25].

$$F_G = B_G \times P_{G-rated} + A_G \times P_{G-out} \quad (3)$$

where $P_{G-rated}$ represents the nominal power of the diesel generator, P_{G-out} ; the output power, while A_G and B_G represent the coefficients of the fuel consumption curve defined by the user (Liter/kWh).

2.8. Battery

The battery storage capacity is given by equation (4) [26]:

$$C_{wh} = (E_L \times AD) / (\eta_{inv} \times \eta_{batt} \times DOD) \quad (4)$$

where E_L represents the daily average load, AD; the number of autonomy days, η_{inv} and η_{batt} are respectively the battery and the inverter efficiency, and DOD is the battery's depth of discharge.

2.9. HOMER simulation

Many design methods exist in the literature of the hybrid energy sizing: hybrid techniques, classical techniques, metaheuristic techniques and software tools. In Ref. [5], an improved marine predators metaheuristic algorithm was used for the sizing of a combined heat/power with PV/Wind/Storage system. A metaheuristic method was also considered to handle the problem of voltage fluctuation in PV system [27]. A hybrid technique of world cup optimization combined with fluid search optimization was proposed for the identification of the parameter of proton exchange membrane fuel cell [28]. In this work the PV/Wind/Diesel/Battery systems are simulated in the 16 un-electrified isolated regions of Chad to determine the optimal systems in terms of costs using the HOMER software. Each region is assumed to have communities that are similar to the three load profiles obtained from Ref. [3]. The HOMER tool has been used extensively for the achievement of similar objectives in the literature [29]. In Ref. [30], some off grid options were simulated to propose optimal hybrid system configurations for the electrification of isolated sites in Cameroon. In Ref. [31], a PV/Wind/Battery/Gas-turbine was designed in HOMER to supply a representative load profile of a village in Ngaoundere. In Ref. [32], a hybrid renewable energy feasibility for the supply of a large resort center located in Malaysia was evaluated with HOMER. Some other works in which the HOMER software has been used are those of Yimen et al. [33] where the PV/Biogas was considered, Iskanderani et al. [34] where the PV/Diesel/Battery was considered, Aberilla et al. [35] where the PV/Wind system was designed and Djiela et al. where PV/Diesel/Battery system was proposed [23].

2.10. Economic evaluation criteria

The HOMER software optimizes costs on several bases, the criteria for optimization are: the net present cost, levelized energy cost, renewable energy rate and finally the quantity of CO₂ that can be released by the optimal system proposed by HOMER.

2.10.1. Net present cost (NPC)

The NPC is the summation of all expenses incurred by the program during the system lifetime, minus the revenue earned [36]. It includes the initial capital cost, replacement cost, operating and maintenance expenses, fuel cost, etc. and is given by equation (5) [37]:

$$Netpresent\ cos\ t = \frac{Cannual.Tot}{CRF(i, lproject)} \quad (5)$$

Where, Cannual. Tot is the annualized cost; CRF represents the capital recovery factor; i is the interest rate and lproject is the project life time.

2.10.2. Cost of energy (COE)

The COE is the ratio between the summation of all costs accumulated during the project lifetime to the electricity produced during the lifetime [38]. In this work the inflation rate is 3.8%. The COE is calculated by equation (6):

$$COE = \frac{Cannual.Tot}{Eprimary.AC + Eprimary.DC + Egridsales} \quad (6)$$

Where, Eprimary AC is the Primary AC load served; Eprimary DC is the Primary DC load served; Egridsales is absolute grid sales.

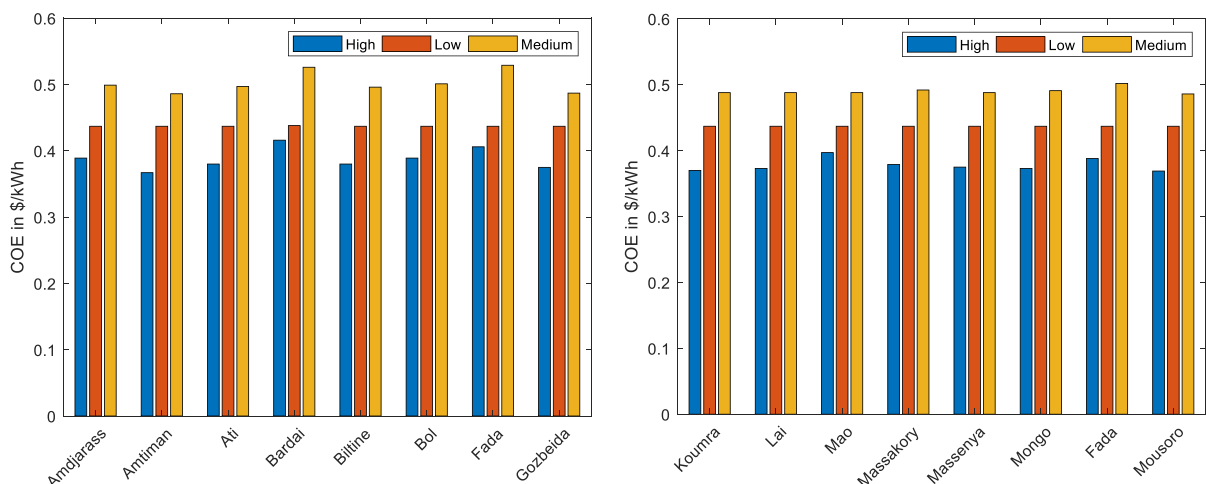


Fig. 10. a) Comparison of COE of energy the first eight sites of study, b).Comparison of COE of energy the others eight sites of study.

2.10.3. Renewable fraction (RF)

The ratio of energy produced by renewable sources to the total amount of energy produced by the entire system is called renewable fraction [39]. Its formula is given by equation (7) [37]:

$$RF(\%) = \left(1 - \frac{\sum P_{diesel}}{\sum P_{generated}} \right) \times 100 \quad (7)$$

3. Results and discussions

3.1. Optimization results

3.1.1. Simulation results

The results show the optimal configuration for each region and different load profiles. The project lifetime is 20 years. The solutions consist of PV/Wind/Diesel/Battery that have the minimum net present cost, cost of energy and CO₂ emission for each type of consumer per site. The optimization of the scenario is based on the different values in each region with the different types of profiles. The penetration rate of renewable energies is a necessary criterion to decision-makers of the energy sector. The results of the different configurations are visible in tables (3 to 18).

In Fig. 10a and b, it is observed that the COE of proposed configurations for the low community consumer are almost identical for all the 16 un-electrified vary from 0.437 to 0.438 US\$/kWh (Tables 3–18). These values of the COE are comparable to those of the

Table 3
Optimization results by type of Amdjarass.

Amdjarass	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	356	0	10	1284	148	LF	0.389	2,52 M	439	1,63 M	99.2	4595
Low	PV/Diesel/Bat	0.151	0	10	3	2.21	CC	0.437	105854	549	8589	0	15666
Medium	PV/Diesel/Bat	178	0	10	905	165	LF	0.499	1,84 M	406	991724	96.8	8206

Table 4
Optimization results by type of Amtiman.

Amtiman	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	332	0	10	1159	163	LF	0.367	2,38 M	2.40	1,53 M	100	58,6
Low	PV/Diesel/Bat	0.143	0	10	3	2.2	CC	0.437	105873	550	8579	0	15669
Medium	PV/Diesel/Bat	166	0	10	839	166	LF	0.486	1,79 M	540	935238	95.5	11358

Table 5
Optimization results by type of Ati.

Ati	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	353	0	0	1222	159	CC	0.380	2,46 M	–	1,61 M	100	0
Low	PV/Diesel/Bat	0.157	0	10	3	2.20	CC	0.437	105861	549	8601	0	15665
Medium	PV/Wind/Diesel/Bat	175	1	10	865	168	CC	0.497	1,83 M	440	978281	96.5	8976

Table 6
Optimization results by type of Bardai.

Bardai	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Wind/Diesel/Bat	355	1	10	1481	157	CC	0.416	2,69 M	646	1,70 M	97,6	11387
Low	PV/Diesel/Bat	0.157	0	10	3	2.21	CC	0.438	105932	551	8604	0	15674
Medium	PV/Wind/Diesel/Bat	209	1	10	936	168	CC	0.526	1,94 M	360	1,10 M	97.3	7097

Table 7

Optimization results by type of Biltine.

Biltine	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	353	0	0	1222	159	CC	0.380	2,46 M	–	1,61 M	100	0
Low	PV/Diesel/Bat	0.133	0	10	3	2.20	CC	0.437	105839	550	8530	0	15672
Medium	PV/Diesel/Bat	117	0	10	901	167	CC	0.496	1.83 M	444	973019	96.4	9095

Table 8

Optimization results by type of Bol.

Bol	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	356	0	10	1284	148	LF	0.389	2,52 M	439	1,63 M	99.2	4595
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	150902	550	8579	0	15673
Medium	PV/Wind/Diesel/Bat	117	1	10	893	165	CC	0.501	1,85 M	448	990395	96.4	9183

Table 9

Optimization results by type of Fada.

Fada	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Wind/Diesel/Bat	331	1	10	1473	158	CC	0.406	2,63 M	692	1,63 M	97.3	12579
Low	PV/Diesel/Bat	0.145	0	10	3	2.20	CC	0.437	105898	551	8563	0	15675
Medium	PV/Wind/Diesel/Bat	233	1	10	832	170	CC	0.529	1.95 M	265	1.14 M	98.2	4805

Table 10

Optimization results by type of GozBeida.

GozBeida	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	342	0	10	1175	165	LF	0.375	2,43 M	2.10	1,58 M	100	55.5
Low	PV/Diesel/Bat	0.131	0	10	3	2.22	CC	0.437	105862	550	8546	0	15672
Medium	PV/Diesel/Bat	182	0	10	813	165	CC	0.487	1,80 M	369	974106	97.2	7289

Table 11

Optimization results by type of Koumra.

Koumra	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	350	0	0	1112	162	CC	0.370	2,40 M	–	1,56 M	100	0
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105880	550	8579	0	15670
Medium	PV/Diesel/Bat	189	0	10	773	165	CC	0.488	1.80 M	394	982574	96.9	7881

Table 12

Optimization results by type of Lai.

Lai	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	336	0	0	1222	161	CC	0.373	2.42 M	–	1.56 M	100	0
Low	PV/Diesel/Bat	0.144	0	10	3	2.21	CC	0.437	105867	550	8574	0	15669
Medium	PV/Diesel/Bat	189	0	10	773	165	CC	0.488	1.80 M	395	982574	96.9	7877

Table 13
Optimization results by type of Mao.

Mao	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	378	0	0	1285	164	CC	0.397	2,57 M	–	1,71	100	0
Low	PV/Diesel/Bat	0.133	0	10	3	2.20	CC	0.437	105879	551	8528	0	15677
Medium	PV/Diesel/Bat	189	0	10	773	165	CC	0.488	1,80 M	395	982574	96.9	7877

Table 14
Optimization results by type of Massakory.

Massakory	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	343	0	10	1246	163	LF	0.379	2,45 M	1,80	1,59 M	100	46.8
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105874	550	8579	0	15669
Medium	PV/Diesel/Bat	182	0	10	833	165	CC	0.492	1,81	417	979561	96.7	8433

Table 15
Optimization results by type of Massenya.

Massenya	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Bat	324	0	0	1302	164	CC	0.375	2,43 M	–	1,55	100	0
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105871	550	15668	0	15668
Medium	PV/Diesel/Bat	178	0	10	801	174	CC	0.488	1,80 M	406	968303	96,8	8160

Table 16
Optimization results by type of Mongo.

Mongo	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	333	0	10	1238	160	LF	0.373	2,42 M	2,10	1,56 M	100	49,9
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105872	550	7188	0	15669
Medium	PV/Wind/Diesel/Bat	185	1	10	801	167	CC	0.491	1,81 M	365	1,81 M	97.2	7175

Table 17
Optimization results of Moussoro.

Moussoro	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Diesel/Bat	353	0	10	1283	152	LF	0.388	2,51 M	439	1,63	99.2	4595
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105887	550	8579	0	15671
Medium	PV/Wind/Diesel/Bat	170	1	10	920	168	CC	0.502	1,85 M	470	981487	96.2	9737

Table 18
Optimization results by type of Pala.

Pala	Architecture							Costs				System	
	Scenario	PV(KW)	Wind	Diesel (Kw)	Battery	converter	Dispatch	COE (US\$/kWh)	NPC(US\$)	O&M Cost (US\$/yr)	Initial Capital (US\$)	RF (%)	CO ₂ (kg/yr)
High	PV/Wind/Diesel/Bat	332	1	10	1159	163	LF	0.369	2,39 M	2,40	1,54 M	100	58,6
Low	PV/Diesel/Bat	0.143	0	10	3	2.22	CC	0.437	105874	550	8579	0	15669
Medium	PV/Diesel/Bat	189	0	10	773	165	CC	0.486	1,79 M	356	982574	97.3	6984

Table 19
Comparative studies of COE and CO₂ results.

References	Geographical Context	Method	Hybrid Composition	COE (US\$/kWh)		CO ₂ (kg/yr)
[42]	Iraq	HOMER	PV/Battery	0.238		–
[41]	Malaysia	HOMER	PV/Wind	0.518		–
[43]	Nigeria	HOMER	PV/DIESEL	0.364		311
[44]	Chana	HOMER	PV/Wind/DIESEL/Battery	0.276		
[45]	Morocco in Oujda	HOMER	PV/Wind/Battery	0.375		–
[40]	Morocco	HOMER	PV/Wind/DIESEL/Battery	0.121		
[46]	Comores	HOMER	PV/Wind/DIESEL/Battery	0,2		1311,407
Our results				min	max	
Current work	Low (16 sites)	HOMER	PV/DIESEL/Battery	0.437	0.438	15666–15670
	Medium (16sites)		PV/Wind/Diesel/Battery or PV/Diesel/Battery	0.486	0.529	7289–11358
	High (16 sites)		PV/Battery or PV/Diesel/Battery or PV/Wind/Diesel/Battery	0.367	0.416	0–12579

Table 20
Sensitivity variables.

Sensitivity variables	Values
Inflation rate (%)	3.00 to 3.80
Project life cycle cost (years)	20 to 25

literature presented in Table 19 where it ranges between 0.121 [40] and 0.518 US\$/kWh [41]. The hybrid system proposed for this type of load profile is the PV/Diesel/Battery (Tables 3–18) for all the sites.

It is also observed in Fig. 10a and b that for the medium consumer, the COE of proposed configurations are between 0.486 and 0.529 US\$/kWh for all the 16 un-electrified regions (Tables 3–18). These values are also in the range of the results observed in the literature (Table 19), between 0.121 [40] and 0.518 US\$/kWh [41]. The hybrid system proposed for this type of profile are either PV/Wind/Diesel/Battery or PV/Diesel/Battery (Tables 3–18) for all the sites.

For the case of the high consumer, it is observed in Fig. 10a and the COE of its proposed configurations are between 0.367 and 0.416 US\$/kWh for all the 16 un-electrified regions (Tables 3–18). These values are in agreement with the results of the literature (Table 19), between 0.121 [40] and 0.518 US\$/kWh [41]. The hybrid system proposed for this type of profile are either PV/Battery, PV/Wind/Diesel/Battery or PV/Diesel/Battery (Tables 3–18) for all the sites. Thus, this work may guide the decision makers and investors in the planning of the electrification in similar low community profile. In all the system proposed where the Diesel generator is included, its runtime is reduced since it is not in standalone mode. This may be a contribution to the achievement of goal number 7 of millennium sustainable development.

For the Chadian government to solve the energy crisis, it can attract investors by exploring such type of feasibility study of options to electrify the isolated areas. The renewable energy implementation with hybrid system design can significantly reduce greenhouse gas emissions and increase electricity access rate in Chad. The National Electricity Company generates electricity using only the diesel generators. Some results of COE and CO₂ values obtained in the literature with different parameters are compared with our results in Table 19.

3.1.2. Sensitivity analysis

The sensitivity analysis was carried out taking into account the different values in Table 20; the inflation rate and the project lifetime.

The Diesel/PV/Battery, PV/Battery and Diesel/PV/Wind/Battery systems were found to be the optimal configurations for the supply of the three communities' load profile in all the considered regions. For the high load profile, the COE reduces from 0.408 US\$/kWh to 0.308 US\$/kWh for the region of Amdjarass when the lifetime and the inflation rate increase (Fig. 11a), while the NPC increases from 2.46 million US\$ to 2.83 million US\$. Similar results are obtained in Fig. 11b, c and 12–13. It means that, when the inflation rate and lifetime are high, the NPC increase while the COE reduces. The sensitivity analysis of the high, low and medium load profiles for the Saharan zone are represented in Fig. 12a–c respectively. The COE values of the high, low and medium load profiles for the Sudanese zone are shown in Fig. 13a–c. When the inflation rate varies from 3% to 3.80%, the COE decreases in different regions.

4. Conclusions

In this work, we have examined the techno-economic feasibility of hybrid systems for the provision of electricity in Chad. Three daily load profiles in 16 un-electrified regions of Chad were considered. After simulation of the options in the HOMER software, the optimal configurations were found.

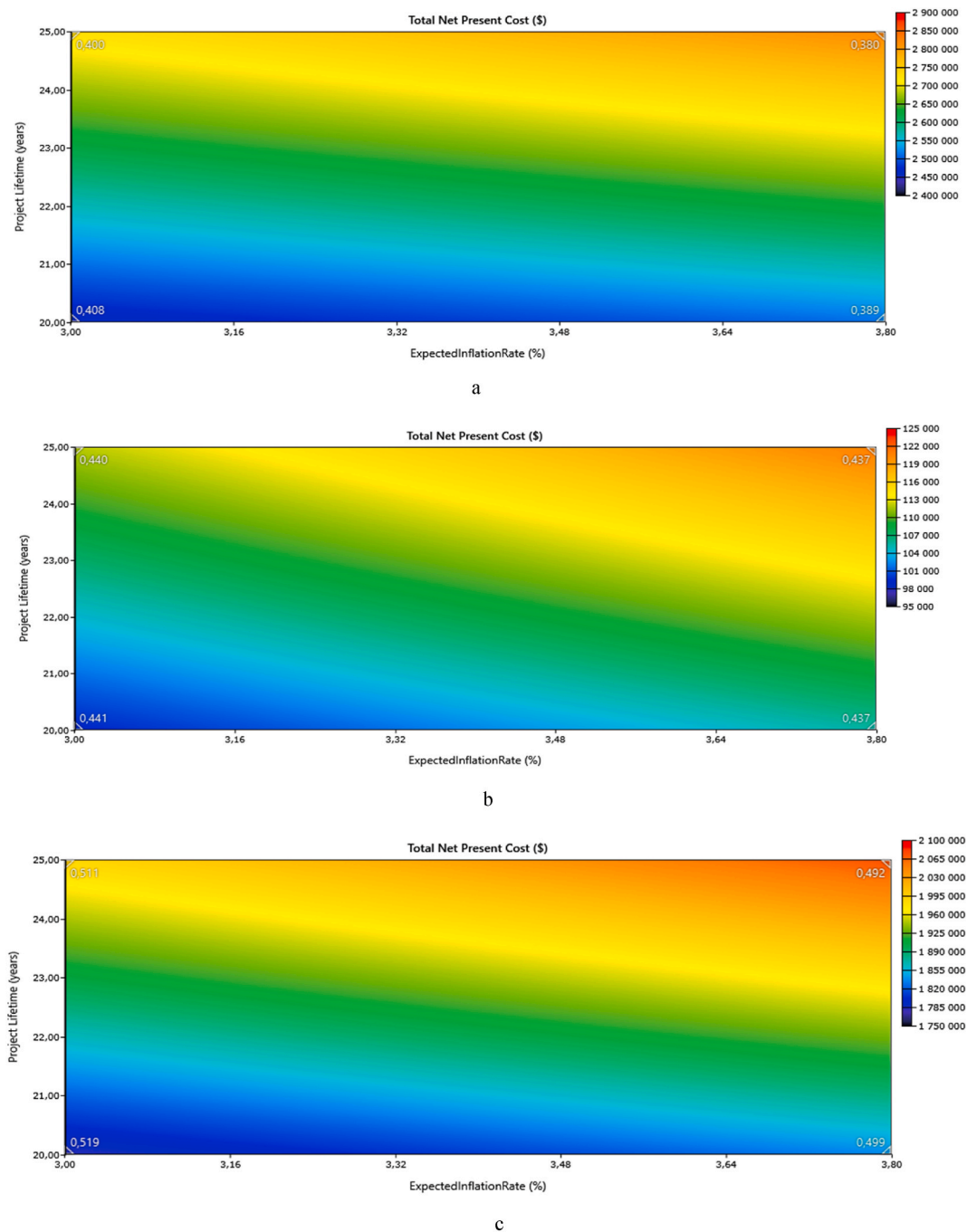


Fig. 11. Sensitivity analysis for Saharan zone: a. High Daily load profile b. Low Daily load profile c. Medium Daily load profile.

For the low community consumer, optimal configurations consist of PV/Diesel/Battery with a COE between 0.437 and 0.438 US \$/kWh and CO₂ emission between 15666 and 15670 kg/year. PV/Wind/Diesel/Battery and PV/Diesel/Battery were the optimal options for the medium community load with COE in the range of 0.486–0.529 US\$/kWh and CO₂ emissions between 7289 and 11358 kg/year. While for the high community load profile, PV/Battery, PV/Diesel/Battery and PV/Wind/Diesel/Battery were optimal with COE between 0.367 and 0.416 US\$/kWh and CO₂ emission between 0 and 12579 kg/year.

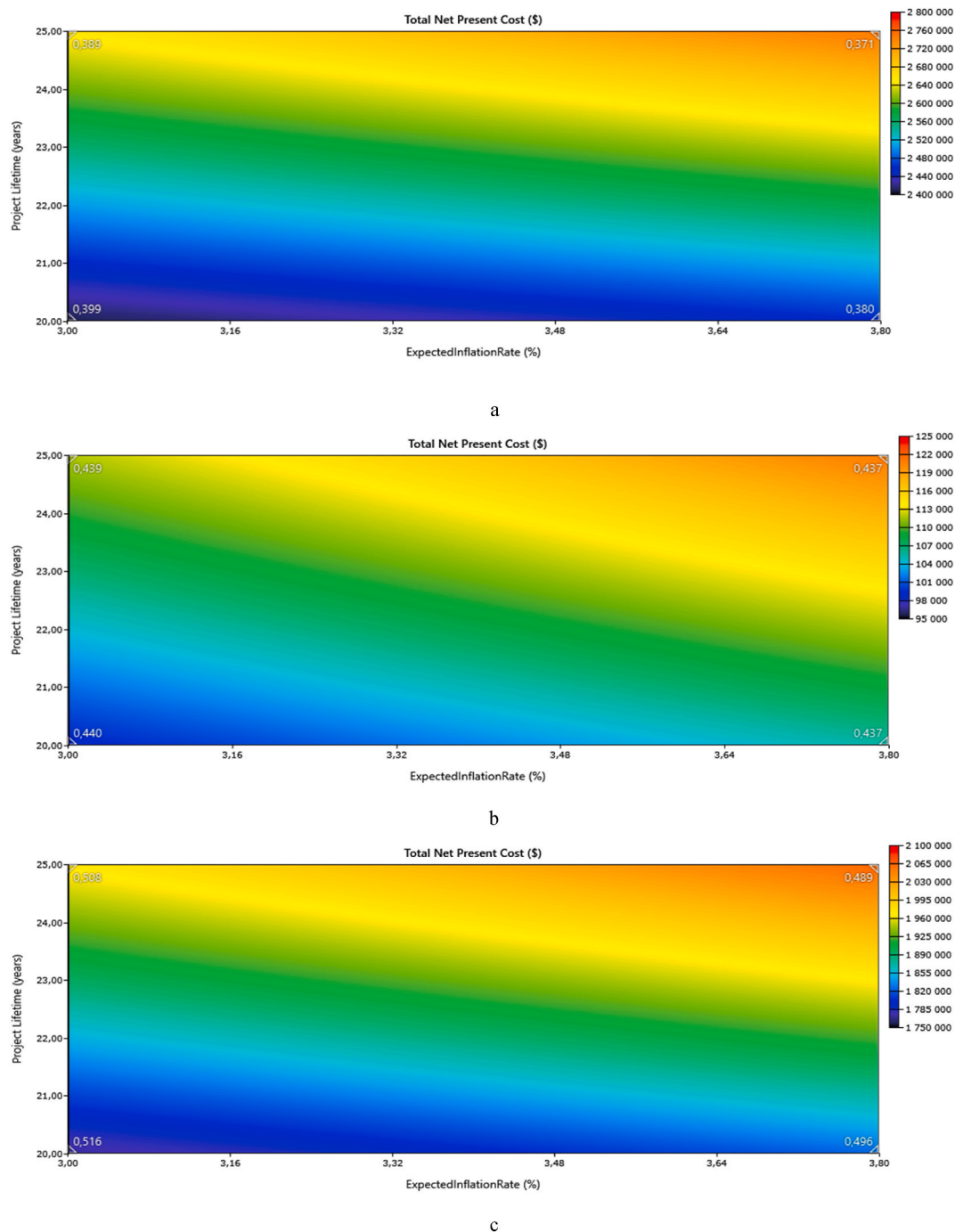


Fig. 12. Sensitivity analysis for Sahelian zone: a. High Daily load profile b. Low Daily load profile c. Medium Daily load profile.

It was observed that, the COE of these proposed configurations were between 0.367 and 0.529 US\$/kWh, indicating that for some sites, it was less than the production cost of electricity in Chad (0.400 US\$/kWh) and therefore profitable. Using the proposed hybrid configurations rather than single diesel generator will also reduce the CO₂ emissions. Sensitivity analysis based on the inflation rate and the project lifetime shows that, when these parameters increase, the COE reduce while the NPC increase. This work may guide decision-makers and investors in the planning and implementation of appropriate energy policies to increase the access rate of

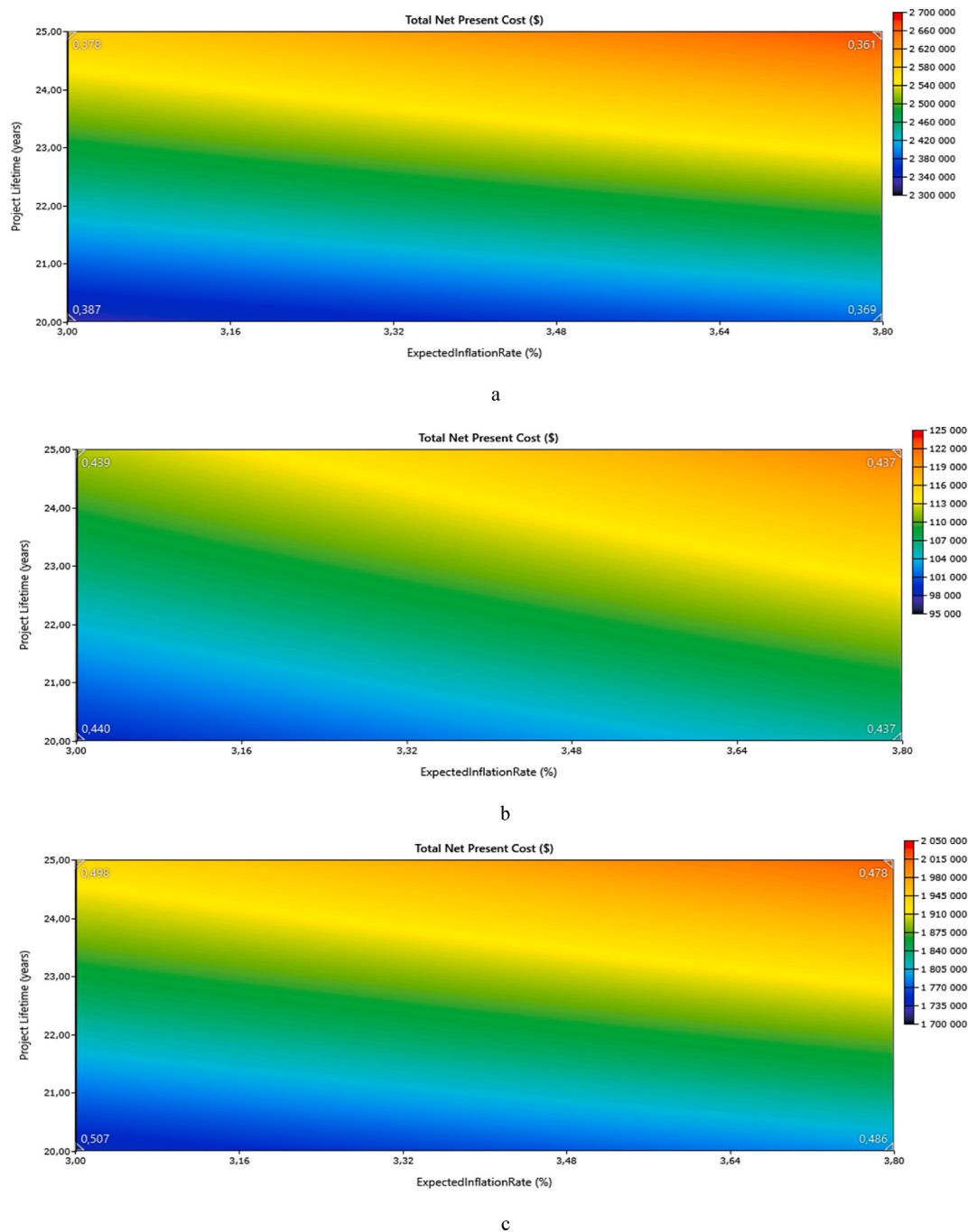


Fig. 13. Sensitivity analysis for Soudanian zone: a. High Daily load profile b. Low Daily load profile c. Medium Daily load profile.

electricity in Chad, especially in remote locations. Future works may investigate the consideration of other sources of energy like hydroelectricity, geothermal power, and the use of fuel cell in hybrid system in Chad. In addition, the comparison of design methods of hybrid energy systems in Chad may be investigated.

Author contribution statement

Elodie Kelly, Brigitte Astrid Medjo Nouadje: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Wrote the paper.

Raphael Hermann Tonsie Djiela, Pascaline Tiam Kapen: Performed the experiments; Analyzed and interpreted the data; Wrote the

paper.

Ghislain Tchuen, René Tchinda: Conceived and designed the experiments; Contributed reagents, materials, analysis tools or data.

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Data included in article/supp. material/referenced in article.

Declaration of interest's statement

The authors declare no conflict of interest.

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